Exp-9 Subnetting

Aim:- Implementation of Subnetting in Cisco packet tracer simulation

Introduction:-

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets. Classless subnetting allows us to use subnet masks other than the default class-based masks, making IP addresses more efficient & allowing multiple networks to be created.

Procedure:-

1.) Open cisco packet tracer

2.) Add required routers, switches & PCs

3.) Connect the devices using appropriate cable

4.) Configure the routers with the required IP address & subnet masks.

5.) Configure the router interface using CLI

- enable
- configure terminal
- select the required FastEthernet/Gigabit Ethernet interface
- Assign the IP address & subnet mask
- Use no shutdown to activate the interface.

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6) Configure the switchports in access mode where required.

7) Assign static IP address, subnet masks & default gateways to the PC's

8) Use the /27 subnet mask: 255.255.255.224.

9) Configure the four subnets:-

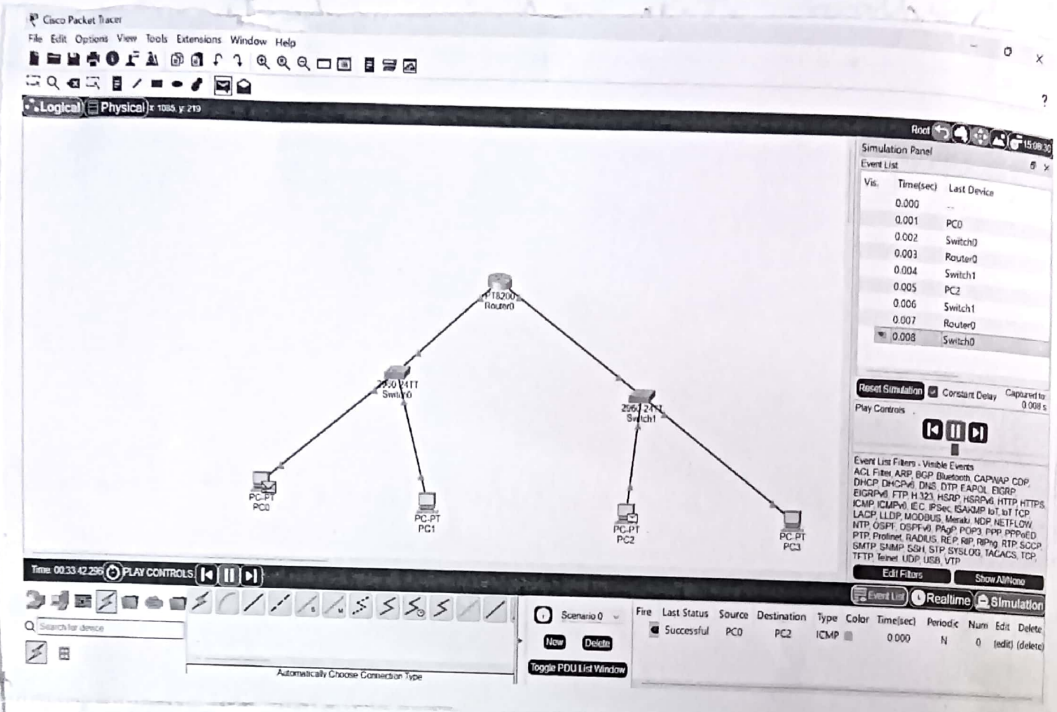
- 192.168.1.0/27
- 192.168.2.0/27
- 192.168.3.0/27
- 192.168.4.0/27

10) Verify the router interfaces & PC's belong to their respective router.

11) Open command prompt on the PC's

12) Use the ping command to test connectivity between PC & respective routers.





Result:-

Thus, subnetting was successfully implemented using CISCO PACKET TRACER.

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